Modal Split and Spillover Effects: Evaluating a General Equilibrium approach to Canadian Urban Transportation Economics

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1 Introduction

Transportation is a key part of everyday life around the world. As Canada faces housing shortages, urban sprawl and a rising cost of living, creating optimal transportation policy is increasingly essential. To achieve long term sustainable economic growth and resiliency requires a comprehensive understanding of urban transportation's many external effects. As such, a general equilibrium approach to urban transportation may be useful.

It is known that transportation has large external costs across many areas. For example, Health Canada has reported that traffic-related air pollution has a country wide social cost estimated at $9.5 billion annually (Health Canada, 2022) in 2015 dollars, and vehicle accident costs have been estimated at $36 billion annually in 2010 dollars (Transport Canada, 2023). As the population grows, and housing continues to spread out from cities, these costs are likely to continue growing unless there is intervention.

Often individual users do not pay the true cost of their mobility choices, causing negative externalities paid by society (Mayeres, Ochelen & Proost, 1996). Schröder et al. (2023) argue that this effectively subsidizes unsustainable modes of transportation. Studies on the monocentric standard urban model have found that cities spread out when transportation is cheaper (Liotta, Viguié & Lepetit, 2022). These links between housing and transportation, especially as Canada faces a housing crisis, make it clear that having properly internalized transportation costs is crucial for a city's sustainable financial outlook.

Since everyone needs to move around in their life, demand for transportation is large and mostly inelastic, generally only changed temporarily by government travel bans such as those seen during the COVID-19 pandemic, or wartime travel restrictions. An increase in work from home can decrease daily commuting, however, Statistics Canada reports that the proportion of Canadian employees in work from home jobs has decreased year over year from 2022 to 2025 (Statistics Canada, 2025).

By understanding the societal costs and benefits of different transportation modes, we aim to find an optimal "modal split". If we can estimate this optimal modal split accurately, then we can create policy which incentivizes transportation users towards this split and create greater societal benefit.

While there is a history of literature which tackles the costs and benefits of various transportation modes, the majority of these studies focus on a particular area such as healthcare or environmental effects. The issue that arises from using the results of these studies to inform policy is that they do not capture the total external costs and benefits created from changes in a population's modal split. Implementing the policy recommendation which addresses one area of externalities may well worsen the externalities in another area of interest.

This links to the theory in economics of general and partial equilibria. What is considered optimal modal split for one area may not be the same as for another. Furthermore, an optimal split across areas for one city may have negative effects on another city, or for the larger province. It can be seen that the problem of optimizing modal split is complex, and many considerations must be made when applying policy recommendations from any particular study.

It follows then, that a comprehensive model which attempts to find a general equilibrium across many cost areas may be useful as the general cost benefit analysis approach may miss large external effects of transportation mode choice.

There is continuing development on computable general equilibrium (CGE) models to address this, though cost benefit analysis approaches are still considered the norm, and urban CGE models have been particularly difficult to transition from theory to application (Robson et al., 2018).

This paper considers existing literature, comparing cost benefit analyses of specific external effect areas which provide partial equilibria estimates, with comprehensive externality models which provide general equilibrium estimates. I consider the benefits and issues surrounding estimating a general equilibrium to optimize modal split, and review how attempts at creating similar models has fared in other areas of economics. Ultimately, I evaluate whether a general equilibrium model should be developed for the Canadian context.

2 An overview of some key cost areas

In this section, I review research on the external costs produced by different modes of transportation in a few key cost areas. Each area has been selected due to the size of its externalities, the amount of literature on it, and its current political importance in Canada at the time of this paper's development.

This is not meant to exhaustively review the totality of external effects caused by transportation mode choice. Instead, it serves to provide evidence of the existence of these external effects being unpriced, causing distortions in the transportation market. A well developed model or framework should include these categories as well as many other areas not addressed in this literature review.

To manage scope, I restrict the number of modes or consolidate some modes together in a wider category. The modes focused on are active mobility (walking or cycling), car travel (mainly gas-powered as these still make up the majority of ownership, though electric vehicles are noted) and public transportation (any form of bus or train). While a complete analysis would involve both costs and benefits, I focus mainly on costs to reduce the scope of the paper.

Due to the interconnected nature of some costs, such as how congestion increases pollution, which is then connected to both environmental costs and healthcare costs, there will be crossover between some of the following sections. This crossover supports the claim that transportation policy requires a comprehensive view across many cost areas.

2.1 Environmental

Extensive research has been done on air pollution caused by various transportation modes. Internal combustion engine vehicles have long been known to contribute significantly to carbon emissions and pollution (Small & Kazimi, 1995). This pollution negatively affects the economy in a variety of ways. Air pollution is known to negatively affect the agriculture industry as well (Dong & Wang, 2023). The healthcare effects of that pollution are touched on in the next section. Analyses comparing gas-powered vehicles and electric vehicles have found considerable support for vehicle electrification to reduce traffic-related air pollution (Ferrero, Alessandrini & Balanzino, 2016).

Furthermore, their widespread use increases both the demand and price of oil and gas, as the transportation sector makes up a large portion of global oil demand (Abrahim et al. 2019). It follows that this increased demand makes investment into the oil industry more attractive and may subsequently make investment in sustainable energy sources less attractive in the short term (Shah, Hiles & Morley, 2018) (Murshed & Tanha, 2021).

2.2 Healthcare

Healthcare makes up a large portion of government spending, and researchers continue to study the relationship between an individual's health outcomes and their choice of transportation. The costs related to users choice of transportation are mainly separated into two areas, accident costs and healthcare costs related to air pollution. In this section, I also consider the health benefits related to active mobility from exercise.

Regarding accident costs, Wu et al. have found that active transportation, such as walking or cycling, improves user health outcomes and reduces their burden on the healthcare system, even when considering the higher accident costs from collisions involving cars and pedestrians (Wu et al., 2021). It has been suggested that by reducing the share of cars in the modal split, we can lower the share of accidents which involve vehicle and pedestrians colliding, therefore lowering the annual total accident costs (Pucher & Buehler, 2016). Researchers have also found increases in accident costs when automobiles are larger and heavier (Anderson & Auffhammer, 2014).

Literature on the healthcare effects of traffic-related air pollution has shown increased risk of morbidity for individuals who drive, commute or live near high traffic areas (Krzyżanowski, 2005). Research has shown that this pollution worsens when traffic congestion worsens, as vehicles spend more time on the road, and pollutants disperse less evenly and become more concentrated (Zhang & Batterman, 2013). The same authors find that this pollution can vary across different road designs.

2.3 Housing, sprawl & taxation

Transportation options are known to have a strong link to land use, urban design, and the location of housing demand and supply (Cervero & Landis, 1995) (Giuliano, Hanson & Agarwal, 2017). In particular, it is important to recognize that transportation has a significant relationship with "sprawl", where new housing supply flows into low density subdivisions far from city hubs which provide the majority of economic activity (Liotta, Viguié & Lepetit, 2022).

It has been found that active mobility choice is heavily affected by trip distance and travel time, both of which increase as housing density decreases and urban sprawl sets in (Ton et al., 2019). Public transportation also becomes less viable as density lowers, as this causes average ridership to lower, and again, increases trip distance and travel time in most locations (Balcombe et al., 2004). The cost of investment also increases as the route becomes longer, as more road or rail infrastructure is needed.

Together, these relationships mean that as sprawl increases, the feasibility of non-car transportation options decreases. Thus, we have this sort of cyclical demand where sprawl incentivizes car ownership and, subsequently, that car ownership supports further urban sprawl as commuters have already taken the sunk cost of vehicle ownership and now look for cheaper homes further from city centers. This supports the idea that there may be a tipping point where the modal split transitions into a new steady state equilibrium which is both unoptimal and hard to exit. This is touched on later in the paper's conclusion.

As transportation options influence urban sprawl, and residents spread out from cities to suburbs, these cities often lose out the tax revenue from those residents, causing them to struggle to maintain the older existing infrastructure within the city (Anas 1983). There are likely some more tax implications related to the type of housing built, as sprawl seems to create a higher share of low-density single family homes due to subdivision zoning laws.

2.4 Takeaways

All of these suggest that there are distortions present in the transportation market. In particular, the literature among almost all cost categories suggests that there is far too much car usage compared to other modes of transportation, which suggests that car travel and its infrastructure is being under priced to consumers and subsequently subsidized by users of other transportation modes.

While these similarly suggest that we should have less car travel in an optimal modal split, they differ on which modes should be increased to replace the lowered share of car travel, and they differ on recommended policy as well. For example, electric vehicles can be seen as a fairly effective replacement to gas-powered vehicles when looking at emissions and the effect of traffic-related air pollution on the environment and healthcare sectors.

However, these changes do not address issues related to the effect cars have on housing and urban sprawl, or the higher accident costs resulting from driver and pedestrian collisions, and policy recommendations for those concerns emphasize shifting towards active mobility and public transport, not towards electric vehicles.

It should be noted that the papers reviewed do not exclusively cover the Canadian urban transportation market, instead often covering U.S. or European cities. However, the Canadian modal split is significantly closer to the U.S. split than Europe's, with both having a much higher share of cars, and the markets are not different enough to suggest that the Canadian market is free of the distortions found in the American or European markets (Pucher, 1988)(Statistics Canada, 2023). This will be discussed later in Section 4.

3 General models review

I now review existing general equilibrium approaches in urban transportation, as well as similar comprehensive approaches in other areas such as healthcare and agriculture.

3.1 Spillover effects & general equilibria

Two useful related economic concepts here are spillover effects and general equilibria.

Spillover effects describe how a change in one area indirectly affects another. These are essentially the effects which we are concerned are missed in the current cost benefit analysis approach, and which make us wary of taking policy recommendations from a study about one cost area in a vacuum.

You can think of a policy maker sort of like a doctor, prescribing medicine (policy) to a patient (the city). If we are not cognizant of the current conditions of the city, then a new policy can have unforeseen effects, or interact with existing policy in a way which defeats its intended purpose. An environmental policy, for example, may appear to reduce pollution directly, but indirectly causes a change in another area which leads to an increase in pollution.

This connects to a question around using partial equilibrium or general equilibrium models in public economics. A partial equilibrium approach may be thought of like that previous environmental example, looking for the optimal level of supply, demand and price within a single market. A general equilibrium approach looks for an equilibrium across several markets which results in each market clearing.

If a government is interested in creating the largest social benefit, then they may be best served looking for a general equilibrium across all the public sectors accounting for spillover effects. This has been supported to be theoretically true, however, this has not always held true in application and there is still ongoing research into the feasibility of applied general equilibrium models (Harberger 1964)(Gohin & Moschini, 2006)(Bolarinwa, 2024).

3.2 General equilibrium models in urban transportation

Schröder et al. in their 2023 paper, "Ending the myth of mobility at zero costs: An external cost analysis," present a comprehensive framework for assessing the external costs of multiple transportation modes simultaneously, covering public transit, motorized individual transport, ride-sharing services, and active mobility across several cost categories at once: air pollution, climate, noise, land use, congestion, accidents, and barrier costs. It also accounts for the health benefits associated with active modes.

They develop a wider framework using data from across Europe and a secondary framework for cities which uses local data to create a model specific to that city. The city of Munich serves as a case study, and the framework is applied to calculate total annual external costs by mode.

The strengths of Schröder et al.'s approach lie in what the partial equilibrium literature reviewed in Section 2 cannot offer: a consistent, unified account of externalities across both modes and cost categories. This reduces the risk that policy recommendations taken from any single cost area will inadvertently worsen outcomes in another. Its structure also makes it adaptive, capable of simulating the effects of proposed policy changes rather than merely describing the current environment and proposing policy.

However, the model has limitations for the Canadian context. It was developed and calibrated for Munich, a dense European city with a modal split, built environment, and transportation infrastructure that differ substantially from any major Canadian city. As will be discussed in Section 4, Canada's lower population density, drastically different current modal split and existing built infrastructure mean that both the existing externalities and the switching costs between modes are likely to be quite different from the European context.

Robson, Wijayaratna, and Dixit's 2018 paper, "A Review of Computable General Equilibrium Models for Transport and Their Applications in Appraisal," provides a robust review of the development and usage of computable general equilibrium (CGE) models for transportation. The authors identify a fundamental weakness in conventional cost-benefit analysis (CBA) as applied to transportation: by assuming static prices in markets external to transport, standard CBA can only account for externalities in an ad-hoc, case-by-case manner. This is only valid under the restrictive assumption that all markets are perfectly competitive and free of distortions.

When that "distortion free" assumption does not hold, and Robson et al. argue that it does not hold in the transportation market, a claim supported by Section 2 of this paper, then the results of a regular CBA approach will understate the true social cost of transportation mode choices. CGE models, by simulating the behaviour of households, firms, and governments, account for how changes in one market spillover through all others. They are designed to produce estimates that avoid the double counting issue present when simply aggregating the results from single cost area models.

Robson et al. are careful to note that CGE models are not without their own difficulties. Data requirements can be prohibitive and the calibration of a CGE model typically involves a "benchmark" data set which can be very difficult to acquire. The authors note that the development of urban CGE models have been constructed mostly using "theoretical or synthesized data," and it has been difficult to implement CGE models for urban transportation. Their review finds that at the time of writing in 2018 only the Netherlands had formal guidelines for incorporating CGE models into transport appraisal, suggesting the method remains underutilized in applied policy contexts despite its theoretical appeal.

Overall, Robson et al. create a compelling evidence based argument against the use of a standard cost benefit analysis for transportation policy due to the large distortions in the urban transportation market. They demonstrate that CGE modelling is theoretically appropriate for capturing the general equilibrium effects that any such model needs to account for, while also identifying practical barriers to implementation, and the lack of current widespread adoption.

Schroder et al. show that it is feasible to develop frameworks similar to computable general equilibrium (CGE) models for use in urban transportation, and while their framework lacks the scope of a true CGE model, it was much more easily constructed while still capturing some of the intended benefits of a CGE model.

3.3 General models in other policy areas

Hendren and Sprung-Keyser's 2020 paper, "A Unified Welfare Analysis of Government Policies," offers a useful point of comparison. While not a model or a framework, it conducts a wide-ranging comprehensive comparison of government policies and their spillover effects across a broad range of public programs. They find that some policies effectively "pay for themselves" by generating positive spillover effects in other areas of the economy, a concept directly relevant to the problems faced when deciding on transportation investment.

As this approach attempts to measure the effects of existing and historical policy, we do not have the same issues around implementing it as we do with computable general equilibrium models. It also has the benefit of being able to track some of the long term effects of policy change, and how these effects can change as the wider economy and transportation market undergo shocks and equilibrium shifts, something which may be a gap in models such as the one proposed by Schröder et al. (2023).

Gohin and Moschini's 2006 paper, "Evaluating the Market and Welfare Impacts of Agricultural Policies in Developed Countries: Comparison of Partial and General Equilibrium Measures," offers a methodological comparison between general equilibrium (GE) models and partial equilibrium (PE) models.

The authors compare three models of agricultural policy, two of which are GE models with the other being a PE model, and find that in practice, partial and general equilibrium approaches produce broadly similar results in the agricultural market. Welfare estimates differ somewhat between approaches, though the authors suggest that spillover effects are generally negligible unless market distortions are large.

As reviewed in Section 2, transportation is a sector characterized by many substantial distortions. It follows that the case for a general equilibrium approach is stronger in transportation than it may be in agriculture, because the distortions are larger and more varied.

These papers suggest that comprehensive welfare analysis is methodologically feasible and has precedent in serious academic work. The success of Hendren and Sprung-Keyser's framework in healthcare policy, and the conditional support Gohin and Moschini provide for general equilibrium approaches in the presence of large distortions, both lend credibility to a comprehensive general equilibrium approach for urban transportation in Canada.

4 Differences within the Canadian transportation context

While there is an increasing amount of research done on computable general equilibrium models for urban transportation, a majority of this research uses

European data and is built for the European context, and relatively few focus on the Canadian context or even the broader North American context.

This gap is significant due to fundamental differences in Canada's geographic environment, its built environment for transportation, and its current modal split.

Canada's modal split is significantly different from the average European country, with a much higher share of car usage in Canada, and a modal split that is closer to America's (Pucher, 1988)(Statistics Canada, 2023). The modal split has been car dominated for at least the last 30 years. Because of this, Canada may be in a local steady state equilibrium with much higher switching costs than European governments face, due to the lack of current infrastructure and the subsequently higher costs of building out infrastructure due to having an underdeveloped construction sector for these types of projects.

Furthermore, the investment that car owners have already made into their vehicle may disincentivize them from switching; as was discussed in Section 2.3, high levels of car adoption feeds into urban sprawl, and the distance of their home from a city center increases the cost of alternative modes to them which subsequently makes car travel more attractive.

These effects make it more difficult to get users to switch away from cars as their primary mode of transportation once sprawl has occurred, and it becomes harder to realistically build new non-car infrastructure due to a lack of urban density and ridership. It should also be noted that there may be some behavioural effects related to the attachment people have to their vehicle and their identity, which may differ from European beliefs (Rau & Matern, 2024).

Subsequently, North America has been slower to adapt general equilibrium concepts into their transportation systems compared to European countries. Development or adaptation of models for the Canadian urban transportation context will not only adjust for these differences, it will better present the issue of optimizing modal split to local policy makers.

5 Conclusion

It is likely that Canada currently has an unoptimized modal split, and could therefore benefit from a general equilibrium approach to urban transportation. The success of similar approaches in other public policy areas, the potential benefits for both transportation researchers and policy makers, the considerable size of spillover effects across cost areas make a general equilibrium approach attractive for optimizing modal split.

However, it should be noted that the creation of a computable general equilibrium model is not without its own difficulties. It requires a lot of data, and a lot of data management. Failure to include enough information around existing externalities could minimize the size of the market's distortions, and therefore cause the general equilibrium model to function similarly to existing partial equilibrium models despite all the extra work in its development.

There are also remaining questions around what level of governance this general equilibrium should focus on. A general equilibrium model may be developed for each city in a province, but this may not result in an optimal split for the overall province. This requires further research to understand.

Before proposing direct follow up research, I would like to mention some related but out of scope research questions which this paper presented me with. As mentioned previously in the housing and sprawl review, modal split being unoptimized for long enough may lead to increasing distortions and an even more unoptimized split. There may even exist a tipping point which shifts the equilibrium into a modal split which is both unoptimal and hard to exit. Research around the long-run risks of unoptimized modal split and the built infrastructure it incentivizes should be conducted.

Related to this potential long-run risk is how policy makers may be able change users transportation habits. In specific, when and how are transportation habits formed? If policy makers can understand this they may be able to target certain demographics with specific policy, such as subsidizing forms of transportation for youth to try and develop life long transportation habits. This could be seen as similar to how modern tech companies will under price their product to gain market share, then increase the price once users are integrated into their product's ecosystem.

I now propose a few follow up projects to build off of this paper. First, a general framework for capturing transportation externalities by mode for cities, allowing any city interested in a general equilibrium model to develop their own from there. Secondly, a general national wide model using transportation data from across Canada. This would provide a rough idea for policy makers in any area even without developing their own model, and would also provide context for those cities developing their own model. Together, these would function similarly to Schröder et al.'s 2023 model reviewed in Section 3.

I also project an estimated lower bound on total external social cost by transportation mode for Canada's current modal split. While the data required for this calculation already exists, we must be careful not to double count any costs. This lower bound on the cost would help communicate the importance and the size of the total external social cost to people outside of the research area.

Lastly, a comprehensive review of historical urban transportation policy subjected to exhaustive cost benefit analysis similar to Hendren & Sprung-Keyser's 2020 paper reviewing historical healthcare policy. This is proposed to help identify useful policy for optimizing modal split without the creation of a complete computable general equilibrium model.

References

Albrahim, M., Zahrani, A. A., Arora, A., Dua, R., Fattouh, B., & Sieminski, A. (2019). An overview of key evolutions in the light-duty vehicle sector and their impact on oil demand. *Energy Transitions*, *3*(1), 81–103. https://doi.org/10.1007/s41825-019-00017-7

Anas, A. (1983, April 1). *The Effects of Transportation on the Tax Base and Development of Cities*. Retrieved April 25, 2026, from https://rosap.ntl.bts.gov

Anderson, M. L., & Auffhammer, M. (2014). Pounds That Kill: The External Costs of Vehicle Weight. *The Review of Economic Studies*, *81*(2), 535–571. https://doi.org/10.1093/restud/rdt035

Balcombe, R., Mackett, R., Paulley, N., Preston, J., Shires, J., Titheridge, H., Wardman, M., & White, P. (n.d.). The demand for public transport: A practical guide.

Bolarinwa, T. (2024). Computable General Equilibrium (CGE) Models: A Comprehensive Review and Future Directions. *MANAGEMENT AND ECONOMICS REVIEW*, *9*(1), 158–167. https://doi.org/10.24818/mer/2024.01-10

Burchell, R. W., Listokin, D., & Phillips, H. (n.d.). The Costs of Sprawl - Revisited.

Canada, H. (2023, May 31). *Traffic-related air pollution*. Retrieved April 8, 2026, from https://www.canada.ca/en/health-canada/services/air-quality/outdoor-pollution-health/traffic-related.html
Last Modified: 2026-02-12.

Canada, T. (2023, January 9). *Statistics on the social costs of collisions in Canada - 2020*. Transport Canada. Retrieved April 28, 2026, from https://tc.canada.ca/en/road-transportation/statistics-data/social-costs-collisions-canada/2020/statistics-social-costs-collisions-canada-2020

Cervero, R., & Landis, J. (1995). The Transportation-Land Use Connection Still Matters. *ACCESS Magazine*, *1*(7), 2–11. Retrieved April 27, 2026, from https://escholarship.org/uc/item/9sg3w7q5

Dong, D., & Wang, J. (2023). Air pollution as a substantial threat to the improvement of agricultural total factor productivity: Global evidence. *Environment International*, *173*, 107842. https://doi.org/10.1016/j.envint.2023.107842

Ferrero, E., Alessandrini, S., & Balanzino, A. (2016). Impact of the electric vehicles on the air pollution from a highway. *Applied Energy*, *169*, 450–459. https://doi.org/10.1016/j.apenergy.2016.01.098

Giuliano, G., & Hanson, S. (2017, March 31). *The Geography of Urban Transportation*. Guilford Publications.

Gohin, A., & Moschini, G. (2006). Evaluating the Market and Welfare Impacts of Agricultural Policies in Developed Countries: Comparison of Partial and General Equilibrium Measures. *Review of Agricultural Economics*, *28*(2), 195–211. Retrieved March 27, 2026, from https://research.ebsco.com/linkprocessor/plink?id=fe3f7844-0089-3a66-bd65-a12e86ea8233

Government of Canada, S. C. (2022, February 9). *Profile table, Census Profile, 2021 Census of Population - Toronto [Census metropolitan area], Ontario*. Retrieved April 28, 2026, from https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/index.cfm?Lang=E
Last Modified: 2023-11-15.

Government of Canada, S. C. (2025, August 26). *The Daily — Number of Canadian commuters increases for fourth straight year in 2025*. Retrieved April 27, 2026, from https://www150.statcan.gc.ca/n1/daily-quotidien/250826/dq250826a-eng.htm
Last Modified: 2025-08-26.

Harberger, A. C. (1964, December 31). Taxation, Resource Allocation, and Welfare. In *Role of Direct and Indirect Taxes in the Federal Reserve System* (pp. 25–80). Princeton University Press. https://doi.org/10.1515/9781400875931-003

Hendren, N., & Sprung-Keyser, B. (2020). A Unified Welfare Analysis of Government Policies*. *The Quarterly Journal of Economics*, *135*(3), 1209–1318. https://doi.org/10.1093/qje/qjaa006

Krzyżanowski, M., & Weltgesundheitsorganisation (Eds.). (2005). *Health effects of transport-related air pollution*. World Health Organization Europe.

Liotta, C., Viguié, V., & Lepetit, Q. (2022). Testing the monocentric standard urban model in a global sample of cities. *Regional Science and Urban Economics*, *97*, 103832. https://doi.org/10.1016/j.regsciurbeco.2022.103832

Mayeres, I., Ochelen, S., & Proost, S. (1996). The marginal external costs of urban transport. *Transportation Research Part D: Transport and*

Environment, *1*(2), 111–130. https://doi.org/10.1016/S1361-9209(96)00006-5
Murshed, M., & Tanha, M. M. (2021). Oil price shocks and renewable energy transition: Empirical evidence from net oil-importing South Asian economies. *Energy, Ecology and Environment*, *6*(3), 183–203. https://doi.org/10.1007/s40974-020-00168-0
Pucher, J. (1988). Urban Travel Behavior as the Outcome of Public Policy: The Example of Modal-Split in Western Europe and North America. *Journal of the American Planning Association*, *54*(4), 509–520. https://doi.org/10.1080/01944368808976677
Pucher, J., & Buehler, R. (2016). Safer Cycling Through Improved Infrastructure. *American Journal of Public Health*, *106*(12), 2089–2091. https://doi.org/10.2105/AJPH.2016.303507
Rau, H., & Matern, A. (2024). Mobility practices in a changing climate: Understanding shifts in car ownership and use across the life course. *Mobilities*, *19*(5), 869–888. https://doi.org/10.1080/17450101.2024.2389846
Robson, E., Wijayaratna, K., & Dixit, V. (2018). A review of computable general equilibrium models for transport and their applications in appraisal. *Transportation Research Part A: Policy and Practice*, *116*, 31–53. https://doi.org/10.1016/j.tra.2018.06.003
Schroder, D., Kirn, L., Kinigadner, J., Loder, A., Blum, P., & Xu, Y. (2023). Ending the Myth of Mobility at Zero Costs: An External Cost Analysis. *Research in Transportation Economics*, *97*. https://doi.org/10.1016/j.retrec.2022.101246
Shah, I. H., Hiles, C., & Morley, B. (2018). How do oil prices, macroeconomic factors and policies affect the market for renewable energy? *Applied Energy*, *215*, 87–97. https://doi.org/10.1016/j.apenergy.2018.01.084
Small, K. A., & Kazimi, C. (1995). On the Costs of Air Pollution from Motor Vehicles. *Journal of Transport Economics and Policy*, *29*(1), 7–32. Retrieved April 25, 2026, from https://www.jstor.org/stable/20053058
Ton, D., Duives, D. C., Cats, O., Hoogendoorn-Lanser, S., & Hoogendoorn, S. P. (2019). Cycling or walking? Determinants of mode choice in the Netherlands. *Transportation Research Part A: Policy and Practice*, *123*, 7–23. https://doi.org/10.1016/j.tra.2018.08.023
Wu, J., Wang, B., Wang, R., Ta, N., & Chai, Y. (2021). Active travel and the built environment: A theoretical model and multidimensional evidence. *Transportation Research Part D: Transport and Environment*, *100*, 103029. https://doi.org/10.1016/j.trd.2021.103029

Zhang, K., & Batterman, S. (2013). Air pollution and health risks due to vehicle traffic. *Science of The Total Environment*, *450–451*, 307–316. https://doi.org/10.1016/j.scitotenv.2013.01.074